

ELIJAH KWAKU ADUTWUM BOATENG

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EDUCATION

Ashesi University MPhil, Intelligent Computer Systems	Berekuso, Ghana Expected 2027
Ashesi University BSc, Computer Science	Berekuso, Ghana 2023

HONORS AND AWARDS

Best Computer Science Thesis Award, <i>Ashesi University</i>	Jun 2023
1 st Place, Career Fair Programming Challenge (Freshman & Senior year), <i>Ashesi University</i>	Mar 2020, 2023
Silver Medallist 2022 IGEN Competition, <i>IGEM Foundation</i>	Oct 2022
Grant Recipient, IGEN Impact Grant (Team Ashesi), <i>IGEM Foundation</i>	Jul 2022
Grant Recipient, Safety and Biosecurity Grant (Team Ashesi), <i>IGEM Foundation</i>	Jul 2022
Recipient, Mastercard Foundation Scholarship	Sep 2019 - May 2023

WORK EXPERIENCE

Aurum Institute Ghana <i>AI Research Scientist Intern</i>	Accra, Ghana Jun 2026 - Aug 2026
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- Redesigned Aires from a RAG-based prototype into a multilayer agentic clinical decision-support system for bacterial infections, grounded in Ghana's Standard Treatment Guidelines and Essential Medicines List.
- Built an agentic workflow coordinating condition routing, evidence retrieval and ranking, and formulary cost-effectiveness analysis with deterministic checks for contraindications, antimicrobial resistance, drug registration and insurance coverage.
- Designed and implemented TRACE, an auditable reasoning and prompting framework that enforces citation-grounded recommendations, blocks absolute contraindications and correctly interprets AND/OR logic in combination therapies.
- Proposed an EMR-agnostic integration architecture using a canonical data schema, CDS Hooks and SMART on FHIR, together with a production engineering plan for deployment across health facilities.
- Supervised and mentored a team member by assigning technical tasks, reviewing code and guiding benchmarking and evaluation experiments while contributing to the core system implementation.

Ashesi University, CSIS Department <i>Research Assistant</i>	Berekuso, Ghana Sep 2024 - Aug 2025
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Supervisors: Dr. Ayorkor Korsah, Dr. Govindha Yeluripati & Mr. Sampson Asare

- Designed and implemented an end-to-end AI-assisted grading pipeline combining OCR with multiple LLM APIs (including OpenAI, Anthropic, DeepSeek and Grok) to evaluate handwritten and descriptive examination scripts.
- Processed and analysed over 1,200 scanned scripts across Algorithms, Data Structures, Artificial Intelligence, Calculus and Discrete Structures, examining grading consistency, fairness and ethical concerns.
- Conducted faculty and student surveys and focus groups on trust, transparency and fairness in AI-assisted grading; incorporated the findings into the pipeline design and contributed to a conference paper.

Aurum Institute Ghana <i>AI Engineer (Contract)</i>	Remote Apr 2024 - Jun 2024
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Supervisor: Nana Kofi Quakyi, MPH

- Developed a RAG-based clinical decision-support system using Ghana's Standard Treatment Guidelines to recommend treatments for 40 high-priority infectious diseases.
- Evaluated the system using 5,000 anonymized patient records and integrated it with hospital management systems for experimental testing.
- Achieved approximately 92% guideline compliance, increased treatment adherence by over 20% and reduced recommendation time from seven to two minutes.

Ashesi University <i>Faculty Intern - Applied Calculus & Object-Oriented Programming & Data Structures & Algorithms</i>	Berekuso, Ghana Aug 2023 - Aug 2024
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- Organized tutorial sessions for students to reinforce course concepts and enhance their understanding of the material for topics in java programming and implementation of complex algorithms and mathematics.
- Assessed and graded assignments, quizzes and problem sets, providing constructive feedback to help students improve their performance.
- Collaborated with the course instructor to develop and implement effective teaching strategies, tailoring support to students who needed more attention in the class.

Black Star Brokerage

Intern, Software Engineer

Accra, Ghana

May 2021- Aug 2021

- Built and deployed a financial data and research portal to analyse primary and secondary bond-market transactions, including yields, trading volumes and price movements.
- Cleaned and visualized market data, enabling the firm to track government fiscal targets and activity across a GHS 2.5 billion bond portfolio.

Black Star Advisors

Intern, Software Engineer

Accra, Ghana

May 2021- Aug 2021

- Collaborated with a co-intern to form an analytics team responsible for building a performance, risk and benchmarking attribution system using python streamlit for pension and mutual funds for the asset management division, which has a hundred million dollars in AUM (Assets Under Management).
- This system enabled the organization to compare the performance and associated risk statistics of up to twenty portfolios to each other and against a benchmark.

KrownPay Limited

Remote Intern, User Interface Designer

Australia, Sydney

April 2021- June 2021

- Designed wireframes, mock-ups and user-story flows in Figma and Adobe XD to guide the development of the company's web and mobile applications.

Ashesi Science Engineering Entrepreneurship Design (SEED) Journal

Backend Developer

Berekuso, Ghana

Mar 2020 - Nov 2020

- Built the journal's backend using Python Flask and MySQL, implementing secure authentication, role-based access control and a validation workflow for submitted papers.

RESEARCH & TECHNICAL PROJECTS

StudentOS

Team Project, AI Product Designer and Engineer

2026

- Designed StudentOS, an AI-powered academic planning system that turns course deadlines, study goals, workload constraints, and student reflections into actionable day-to-day learning plans.
- Architected the product around intelligent scheduling, course-level planning, academic journaling, spaced repetition, and agent-driven study recommendations.
- Designed onboarding and planning flows that translate broad academic goals into measurable learning objectives, time constraints, and personalized scheduling preferences.
- Prototyped a unified dashboard combining courses, deadlines, study sessions, reflections, and adaptive recommendations, and explored signals such as deadline proximity, study consistency, task completion, and reflection history to improve recommendation quality.

Unsupervised Speech Recognition with wav2vec-U

Independent Project, NLP Research

2026

- Reproduced the wav2vec-U architecture to study unsupervised speech recognition, mapping self-supervised acoustic representations to phoneme sequences without paired speech-text data.
- Implemented the adversarial learning framework, training a generator to convert wav2vec 2.0 features into phoneme distributions and a discriminator to distinguish generated sequences from phoneme sequences derived from unpaired text.
- Built the complete pipeline for feature extraction, frame segmentation, phoneme vocabulary construction, GAN training and phoneme-level decoding, adapting the implementation for low-resource data and limited compute.

- Improved training stability by tuning representation dimensionality, discriminator capacity and generator-to-discriminator update frequency, then evaluated performance using phoneme error rate, validation perplexity and sequence-level analysis.

Interius: Autonomous Backend Engineering System

Independent Project, AI Engineering

2026

- Built a multi-agent system that converts natural-language specifications into working, multi-file backend applications through requirements analysis, architecture, implementation, review and automated repair stages.
- Developed a sandboxed validation layer that executes generated code, detects runtime failures and triggers automated repairs before packaging successful builds in Docker containers.
- Added a retrieval-augmented chat interface that helps users understand generated codebases and supports documentation, Git integration and deployment to platforms such as Render.
- Shipped the complete workflow as a desktop application, achieving a 90% end-to-end success rate and an 85% first-attempt success rate on backend generation tasks.

Automatic Speech Recognition

Independent Research Engineering Project

2025

- Fine-tuned a pretrained Whisper-Tiny model for Ghanaian-accented, Twi-influenced English ASR using a low-resource subset of the AfriSpeech dataset.
- Trained on 1,315 audio-text pairs with substantial speaker, accent, and pronunciation variability.
- Achieved 34.28% WER and 19.07% CER, substantially outperforming an RNN-CTC baseline of roughly 85% WER.
- Demonstrated effective transfer from pretrained speech representations under severe data constraints, highlighting their potential for low-resource African-accented ASR.

Machine Translation

Independent Research Engineering Project

2025

- Reproduced the Transformer encoder-decoder architecture from the [Attention Is All You Need](#) paper and applied it to English-French machine translation without pretrained models.
- Constructed and filtered a parallel corpus of 130,187 sentence pairs, then implemented scaled dot-product attention, multi-head attention, positional encoding, residual connections and layer normalization from scratch.
- Built and trained the complete sequence-to-sequence pipeline using cross-entropy loss and greedy decoding, achieving a BLEU score of 51.77.
- Conducted qualitative error analysis to examine translation failures and attention behaviour, establishing a controlled baseline for studying Transformer design choices.

Ashesi University, University of Ghana

Research Assistant

Berekuso, Ghana

Jul 2021 - Dec 2022, May 2023 - Aug 2023

Advisors: Dr. Joseph A. Mensah & Dr. Justice K. Appati

- Investigated novel approaches to face recognition under multiple constraints, focusing on occlusion recovery, low-resolution image enhancement, and computational efficiency.
- Implemented and optimized a hybrid DWT-PCA/FFT algorithm, improving recognition accuracy by 15% over standalone methods in occluded-face experiments.
- Conducted comprehensive literature review on multi-constrained face recognition techniques, identifying key research gaps and potential areas for improvement and innovation.
- Setup and run experiments under single and combined constraint settings.
- Developed an interface program to seamlessly integrate MATLAB and Python code bases, enhancing the project's overall efficiency and enabling more complex analyses.
- Created a simulation program to model various occlusion scenarios, allowing for robust testing of the face recognition algorithm under diverse real-world conditions.
- Created infographics, posters and slides for papers and presentations.

International Genetically Engineered Machine (IGEM) Competition

Silver Medallist, Team Ashesi

Paris, France

Oct 2022

Principal Investigator: Dr. Elena Rosca

- Collaborated with a 10-member team to develop a multifunctional biosensor using engineered E. coli cultures to detect arsenic and iron as pathfinder elements for gold prospecting.
- Co-designed three registered composite genetic parts: a UV-inducible kill switch using T4 endolysin to prevent bacterial escape, a Ferric-ion sensor producing a visible pink signal and an Arsenic sensor producing a visible yellow signal.
- Researched the biosensor architecture and identified chromoproteins as a cost-effective alternative to fluorescent proteins for instrument-free visual detection.
- Contributed to the team's successful \$5,000 Safety and Biosecurity Grant application, one of five selected from 48 teams, and co-authored an interdisciplinary article on safe gold prospecting in Ghana.
- Led the development of the team's wiki and collaborated with the University of Patras to translate a synthetic biology comic book into Twi for middle-school education in Ghana.

PUBLICATIONS & RESEARCH

Publication: Joseph A. Mensah, Justice K. Appati, Elijah K. A. Boateng, Eric Ocran, and Louis Asiedu, "FaceNet Recognition Algorithm Subject to Multiple Constraints: Assessment of the Performance," Scientific African, vol. 23, p. e02007, 2024. doi: 10.1016/j.sciaf.2023.e02007.

Thesis: Elijah K.A Boateng, "Face Recognition under Multiple Constraints with Application to Attendance Management," Ashesi University, 2023. Available: <https://hdl.handle.net/20.500.11988/1029>. (*Best Computer Science Thesis*)

Article: AshesiGhana team, "An Affordable & Fast Approach to Gold Prospecting in Ghana," AshesiGhana, 2022. Available: <https://2022.igem.wiki/stony-brook/collaborations/global-journal>.

LEADERSHIP AND CO-CURRICULAR EXPERIENCE

Ashesi Student Council

Academic Chairperson

Berekuso, Ghana

Feb 2022 - Mar 2023

- Launched the "Safe Study Spaces" initiative with the Office of the Provost, introducing music, quiet and creative environments tailored to different student learning preferences.
- Partnered with the Academic Advising Office to develop a GPA forecasting system that helped students set semester targets and track progress toward graduation.
- Represented students on the Academic and Judicial Councils, contributing to academic policy discussions and advocating for fair disciplinary processes.
- Mentored students who later assumed Student Council leadership roles, supporting leadership development and continuity.

Ashesi Math Resource Centre

Lead

Berekuso, Ghana

Jan 2021 - Aug 2022

- Coordinated the Math Resource Centre's operations, matching students with tutors for targeted support across mathematics courses.
- Led group tutorials and one-on-one sessions, helping students understand difficult concepts and strengthen their problem-solving skills.
- Promoted peer learning and student engagement with mathematics through campus initiatives, including Ashesi's Pi Day programme.

Berekuso Community School

Berekuso Math Project, Volunteer Tutor

Berekuso, Ghana

Jan 2021 - Aug 2022

- Tutored 100+ students in mathematics in preparation for high school entrance examinations, contributing to improved mathematics pass rates.

SKILLS AND INTERESTS

- **Programming:** Python, Java, C++, SQL, JavaScript
- **Machine learning:** PyTorch, NumPy, Pandas
- **Tools:** Docker, Git
- **Research Interests:** Multi-agent learning, reliable AI systems, AI for healthcare and machine learning for low-resource environments.